Franziska Hoerbst

Birthday	Email	Web	
22 September 1998	franziska@hoerbst.net	franziskaaaah.github.io	
Address	Telephone	Nationality	Gender
Apfoltern 28, 4261 Rainbach i.M.	+44 7901 68 13 36	Austrian	female

Previous, current and future areas of research

– Bayesian inference
– Information theory
– RNA-Sequencing
– Plant communication
– Feminist epistemology
– Political economy
– Ecological Civilization
– dialectical approaches

Languages in order of proficiency

——— German
——— English
—— Spanish

Research & education

2024-2025	**John Innes Centre**, Norwich Research Park, UK
Postdoc	Computational Systems Biology
2020-2024	**John Innes Centre & University of East Anglia**
PhD	Computational Systems Biology
	Doctoral Thesis supervised by Richard Morris
	—— “Bayesian Inference in the Analysis of RNA-Seq Data”
2017-2020	**University of Vienna**, Austria
BSc	Molecular Biology
	+ Thesis in Computational Biophysics supervised by Bojan Zagrovic at Max Perutz Labs
	—— “Exploring the Physicochemical Foundation behind Travelling Autologous Protein-mRNA Paris in the Phloem of Plants”
-2017	Primary and high school, Austria

Research publications

2022	Exact Bayesian inference for the detection of graft-mobile transcripts from sequencing data
Royal Society Interface	F Hoerbst, M Tomkins, S Gupta, F Apelt, J Kehr, F Kragler, RJ Morris
2023	baymobil: A Python package for detection of graft-mobile mRNA using exact Bayesian inference on RNA-Seq data
Preprint	F Hoerbst, RJ Morris, M Tomkins
2024	A Closed-Form Solution to the 2-Sample Problem for Quantifying Changes in Gene Expression using Bayes Factors
Preprint	F Hoerbst, GS Sidhu, M Tomkins, RJ Morris
2024	Re-analysis of mobile mRNA datasets highlights challenges in the detection of mobile transcripts from short-read RNA-Seq data
Nature Plants	P Paajanen, M Tomkins, F Hoerbst, R Veevers, M Heeney, HR Thomas, F Apelt, E Saplaoura, S Gupta, M Frank, DJ Walther, C Faulkner, J Kehr, F Kragler, RJ Morris + Research briefing: Current RNA-seq evidence of mRNA mobility is largely due to noise in the data
2024	Bayesian Inference in the Analysis of RNA-Seq Data
Thesis	F Hoerbst
2025	On the overinterpretation of mass screening data – the example of mobile mRNA
Trends in Plant Science	F Hoerbst, M Tomkins, RJ Morris
2025	A Bayesian framework for ranking genes based on their statistical evidence for differential expression
Under review	F Hoerbst, GS Sidhu, T Omori, M Tomkins, RJ Morris
2025	What is a differentially expressed gene?
Under review	F Hoerbst, GS Sidhu, M Tomkins, RJ Morris

Science communication, engagements, workshops & teaching

2025	**Movement Research Unit** — Volunteering to do the research that helps movements win
2022-2025	**UEA Biodiversity and Climate Action Network (BCAN)** unite staff and students to build a positive future for our university community and beyond — developing and running workshops and campaigns — contributions to the working group on higher education, with the vision of integrating concepts surrounding global citizenship into all student curricula
2022-2024	**JIC Student Voice (JSV)** — Running campaigns to support the John Innes student community to thrive (e.g. on statistical literacy)
2019-2022 Podcast	**5MinutenClimateChance** Climate science podcast & radio broadcast Co-founder, speaker, journalistic researcher & producer involved in writing, editing, or reviewing of 100+ episodes
2023 JIC	**Undergraduate student supervision** Computational and Systems Biology
2022 UEA	**University teaching assistant** Mathematics for Computer Scientists
2018-2020	**ScienceCenter Network Vienna** Hands-on science communication — Hosting workshops, developing hands-on and tinkering activities for audiences of all ages and backgrounds

Further publications, presentations & conferences

2024	Let's 'do different': Planetary Citizen Education Proposal
Working paper	E Honeybun-Arnolda, P Stephanides, F Hoerbst, N Longhurst, D Maguire, H Hoechner
2024	MaxEnt24: Maximum Entropy
Ghent, Belgium	Conference Talk
	—— Best Presentation Award
2024	ECMTB24: European Conference on Mathematical and Theoretical Biology
Toledo, Spain	Conference Talk
2024	18th Microsymposium on RNA
Vienna, Austria	Poster presentation
2023	Physics of Life
Harrogate, UK	Poster presentation
2023	17th Microsymposium on RNA
Vienna, Austria	Conference Talk
2022	PVB2022: Plant Vascular Biology
Berlin, Germany	Poster presentation with Melissa Tomkins
2021	A day trip through phloem
Blog	Easy read research project description
2023	Life out there? Scanning plant space for new communication networks
Blog	Easy read research paper teaser
2023	Activism and Advocacy as a Researcher
Panel	Open discussion at the Annual Assembly of the Tyndall Centre for Climate Change Research
2022	EMBO Codon usage conference
Edinburgh, Scottland	Poster presentation + flash talk

Prizes, Awards, Accolades

2025	—— **John Innes Foundation Prize for Excellence in Scientific Research**
2023	—— **JIC Director Award** A yearly award given by Director Graham Moore to individuals to acknowledge and honour outstanding scientific and cultural contributions to the John Innes Centre
2022-2024	**JIC Student Voice** — Running campaigns to support the John Innes student community to thrive
	With 5MinutenClimateChance
2020	—— **Austrian SDG Award**
2020	—— **Austrian Environmental Journalism Prize**
2021	—— **Top 5 Salto Awards**
2012-2018	**Radius 106.6, Freies Radio Freistadt, Radio Orange 94.0** Radio presenter, speaker, producer —— Student award by the Austrian Federal Ministry of Education and OE1 for best broadcast 2016
2017 & 2018	**Coordination and execution of charity radio marathons** 4 Tage für den Sozialmarkt, 4 Tage für die Lebenshilfe
2016	—— Regional Movie Award @YOUKI Film Festival
2008-2017	—— Prima La Musica + other music competitions/awards

Contact for references

Professor Richard J. Morris Group leader John Innes Centre, Norwich, UK PhD supervisor Richard.Morris@jic.ac.uk	Dr Melissa Tomkins Senior Researcher John Innes Centre, Norwich, UK PhD supervisor Melissa.Tomkins@jic.ac.uk
Hannah Rae Thomas, PhD Job position: 百人计划研究员 National title: 国家级人才称号 Institute/Department: 蔬菜科学研究所园艺系 Website: Hannah Rae Thomas-浙江大学个人主页 Email: HR.Thomas@zju.edu.cn Phone/wechat: 19967301244	Dr Noel Longhurst Lecturer in Energy & Climate Change School of Environmental Sciences University of East Anglia Founding member of UEA Biodiversity and Climate Action Network (BCAN) N.Longhurst@uea.ac.uk